Questions

Q1.

Energy from the nuclei of atoms can be used in medical diagnosis and treatment.

(i) Fluorine-18 is a radioactive isotope used in PET scanners for medical diagnosis.

Explain why fluorine-18 must be produced close to the hospital where it is used.

(2)

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(ii) Some tumours inside the body can be treated by using either alpha radiation or gamma radiation.

Explain why the source of alpha radiation is usually inside the body but the source of gamma radiation can be outside the body.

(4)

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(Total for question = 6 marks)

Q2.

Radioactive substances are used in the generation of electricity.

State **two** other uses of radioactive substances.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q3.

Radiotherapy can involve irradiation of patients.

Radioactive tracers can involve contamination of patients.

State **two** differences between irradiation and radioactive contamination.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q4.

Figure 1 is a diagram of a nuclear reactor, used in the generation of electricity.

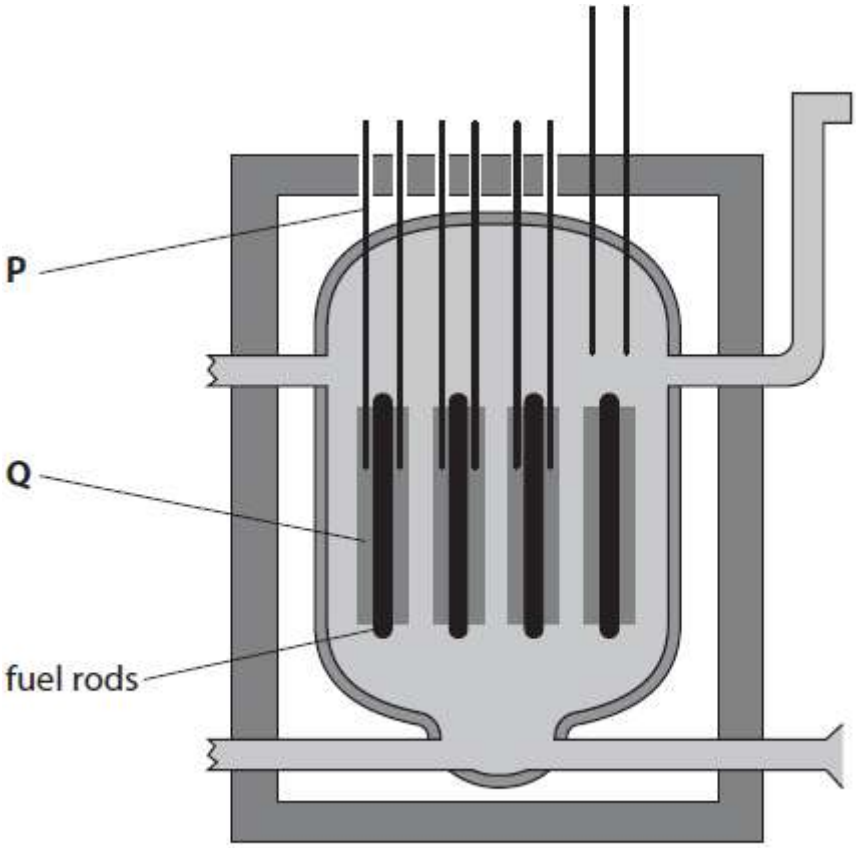

Figure 1

P may be used to shut down the reactor when necessary.
Q slows down neutrons to enable a chain reaction to take place.

State the name of the two parts labelled **P** and **Q**.

(2)

P .....

Q .....

(Total for question = 2 marks)

Q5.

Alpha, beta and gamma are all **ionising** radiations.

Give the meaning of the term **ionising**.

(1)

.....

.....

(Total for question = 1 mark)

Q6.

Gamma radiation is part of the electromagnetic spectrum.

When the nucleus of an atom emits a gamma ray, the number of protons in the nucleus and the number of neutrons in the nucleus **do not** change.

State how the nucleus **does** change when it emits a gamma ray.

(1)

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.....

(Total for question = 1 mark)

Q7.

Figure 6 shows the decay curves of two different isotopes, Q and P.

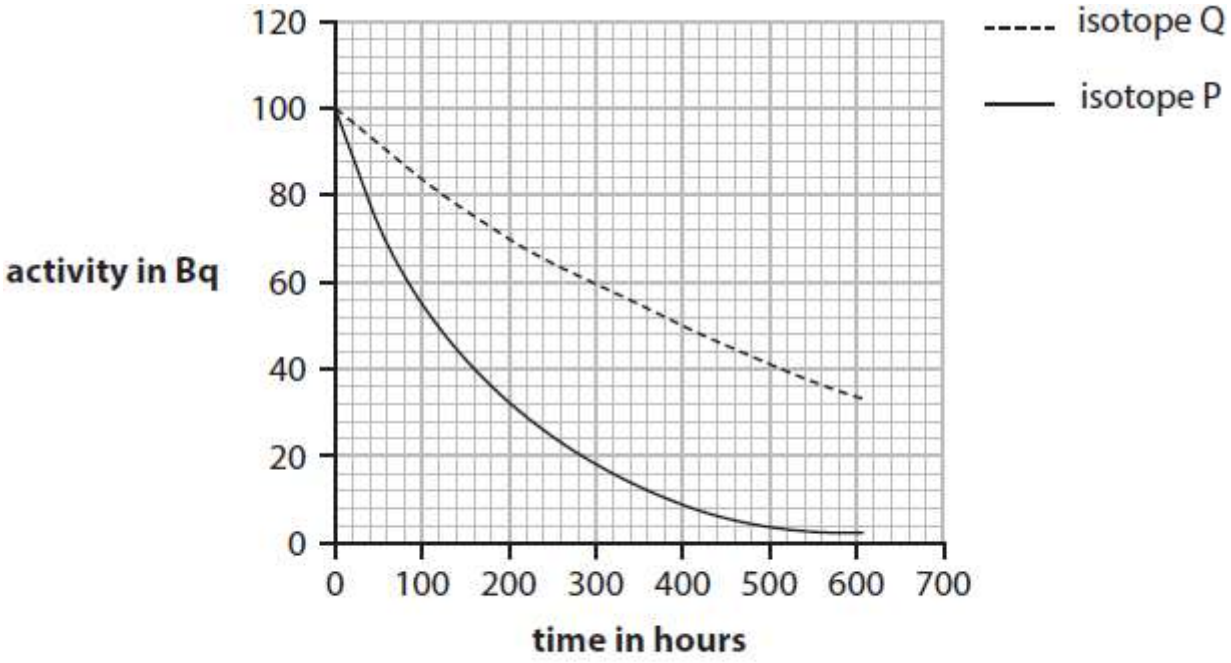

Figure 6

(i) Use the graph in Figure 6 to determine the half-life of isotope P.

(2)

half-life of isotope P = ... hours

(ii) Suggest a reason why the sample of isotope Q could be more dangerous to humans than the sample of isotope P.

(1)

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.....

(Total for question = 3 marks)

Q8.

Figure 4 shows a Geiger-Müller (GM) tube used for measuring radioactivity.

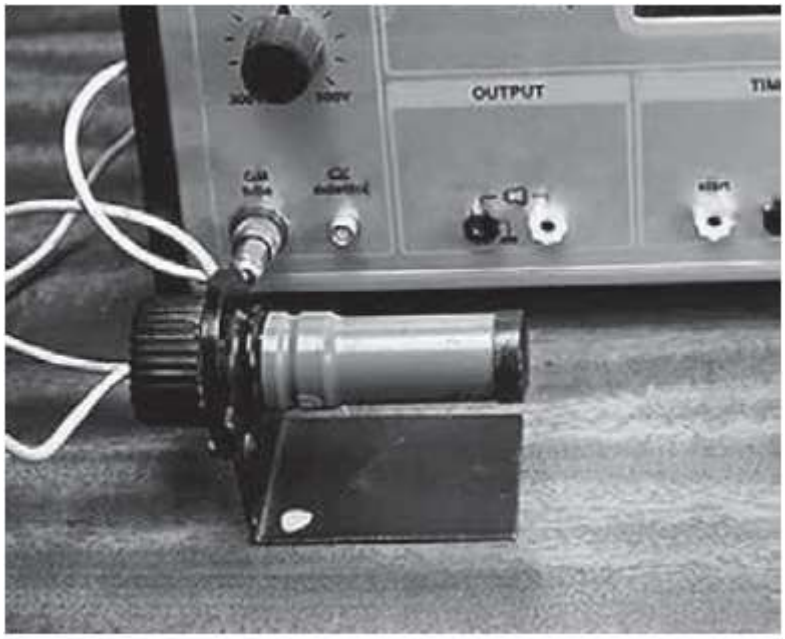

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Figure 4

A hospital uses a radioactive isotope with a half-life of 6 hours.

A technician measures a count rate of 80 counts per minute (cpm) from this isotope.

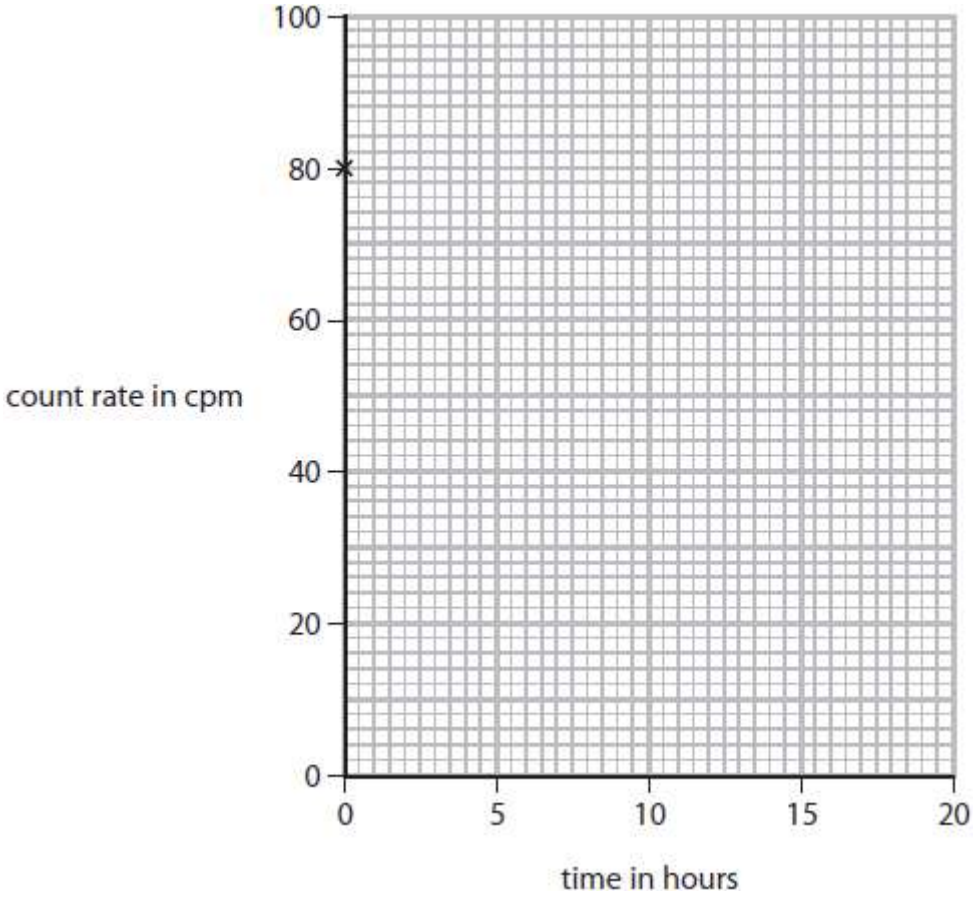

Figure 5

Complete the graph on Figure 5, as accurately as possible, to show how the count-rate from this isotope will change from the time of the first measurement.

The first point is already drawn in Figure 5.

(3)

(Total for question = 3 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	an explanation including: (fluorine-18 has) a short half-life (1) (so) it must be **used** as soon as possible after making (1)	 decays too quickly related to transport / proximity ignore arguments about harm to person / the environment	**(2)**

Question Number	Answer	Additional guidance	Mark
(ii)	an explanation including:		**(4)**
	alpha short range/low penetration (1)	accept highly ionising	
	(so) needs to be close to the tumour (1)		
	gamma long range/high penetration (1)	accept weakly ionising	
	(so) can get into the body from outside (1)	pass through the skin	
		'alpha more ionising than gamma' 1 mark by itself	

Q2.

Question number	Answer	Additional guidance	Mark
	Any two from smoke alarm (1) food irradiation (1) sterilising (1) detecting leaks/cracks (1) gauging thicknesses (1) in medicine diagnostic (1) in medicine therapeutic (1) dating (archaeological samples) (1)	 named named	(2)

Q3.

Question	Answer	Additional guidance	Mark
	any two from :- irradiation does not make an object radioactive (1) irradiation can be stopped by shielding (1) irradiation source is outside the object (1) irradiation stops when the source is removed (1)	 ORA ORA source of contamination is in or on the object effect of contamination is continuous allow you can move away from irradiation	**2** **AO1.1**

Q4.

Question number	Answer	Additional guidance	Mark
	P - control rods (1) Q - graphite/moderator (1)	boron steel rods heavy water	**(2)**

Q5.

Question	Answer	Additional guidance	Mark
	loss / gain of electrons (from atoms)	accept knock electrons out (of atoms)	**1** **AO1.1**

Q6.

	Answer	Additional guidance	Mark
	One from: rearrangement (of particles) (1) loses/emits energy (1) becomes (more) stable (1)		**(1)** **AO1**

Q7.

Question	Answer	Additional guidance	Mark
(i)	correct use of the graph (1)	horizontal line in range 48-52 Bq to line for either isotope allow two horizontal lines showing halving of the activity	**2** **AO2.1**
	110 to 130 (hours) inclusive (1)	award full marks for the correct answer without working	

Question	Answer	Additional guidance	Mark
(ii)	**one** suggestion from: (Q) has longer half-life (1) (Q) does not decay as quickly/ takes longer to decay (1) (Q) (maintains) a higher (level of) activity (1) (Q) gives a higher dose (1) (Q) emits more ionising (type of) radiation (1)	allow reverse arguments activity decreases more slowly (always) more (radio)active / more decays per second	**1** **AO2.2**

Q8.

Question Number	Answer		Mark
	• point after first half-life 6, 40 (1) • point after second half-life 12, 20 (1) • point after third half-life 18, 10 (1)	within 1 small square by eye smooth curve starting at 80, with a decreasing gradient passing through one correct half-life point scores 2 marks smooth curve starting at 80, with a decreasing gradient passing through two correct half-life points scores 3 marks **if no other mark scored** smooth curve showing decreasing gradient but not going through any correct points scores 1 mark	**(3)** AO 3 1a